EISENSTEIN POWERS ON ONE PAGE

ω^η or η^ω: size, angle and 23

Swap two Eisenstein integers: one power is positive real; the other has a phase.

Choose the pair $\omega = e^{i(2\pi/3)} = -\frac{1}{2} + \frac{\sqrt{3}}{2}i, \qquad \eta = 1 + 2\omega = i\sqrt{3} = \sqrt{3}\,e^{i\pi/2}.$

Both lie in $\mathbb{Z}[\omega] = \{a + b\omega : a, b \in \mathbb{Z}\}$. The principal logarithms are

$$\operatorname{Log}\omega = i\frac{2\pi}{3}, \qquad \operatorname{Log}\eta = \frac{1}{2}\ln 3 + i\frac{\pi}{2}.$$

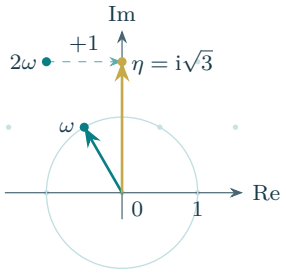
The rule: separate magnitude and phase

Write $z = re^{i\theta} \neq 0$, with principal argument θ, and $w = a + ib$. Then

$$z^w := e^{w \operatorname{Log} z} = \underbrace{e^{a\ln r - b\theta}}_{\text{magnitude}}\,\underbrace{e^{i(b\ln r + a\theta)}}_{\text{phase}}.$$

The result is real exactly when $b\ln r + a\theta \in \pi\mathbb{Z}$. Complex numbers have no usual $<$ ordering, so compare their **magnitudes** and record their angles.

Two points on the lattice



Forward: ω^η is positive real

The unit-circle base has $r = 1$; its logarithm is purely imaginary. So is the exponent:

$$\omega^\eta = e^{(i\sqrt{3})\cdot i(2\pi/3)}$$
$$= e^{-2\pi/\sqrt{3}}.$$

The phase is zero: there is no residual rotation.

$$\boxed{\omega^\eta = e^{-2\pi/\sqrt{3}}}$$
$$\omega^\eta \approx 0.0265799$$

Reverse: η^ω has a phase

Now the logarithm has both a real and an imaginary part:

$$\eta^\omega = e^{\omega(\frac{1}{2}\ln 3 + i\pi/2)}$$
$$= \rho\, e^{i\phi},$$

where

$$\rho = 3^{-1/4}e^{-\pi\sqrt{3}/4} \approx 0.1949519,$$
$$\phi = \frac{\sqrt{3}\ln 3 - \pi}{4} \approx -0.309685 \text{ rad.}$$
$$\eta^\omega \approx 0.185678 - 0.059413i$$

Which has the larger magnitude?

$$\boxed{|\omega^\eta| \approx 0.02658 \;<\; 0.19495 \approx |\eta^\omega|}$$

Both are proved transcendental

Both inputs are algebraic:

$$\omega^2 + \omega + 1 = 0, \qquad \eta^2 + 3 = 0.$$

By Gelfond–Schneider [1], an algebraic base outside $\{0, 1\}$, raised to an algebraic exponent outside $\mathbb{Q}$, is transcendental. Thus **both powers are transcendental on every branch**.

The imaginary part $\sqrt{3}$ of η is irrational.

The arithmetic coda: 23 stays prime

The Eisenstein ring $\mathbb{Z}[\omega]$ is Euclidean, so it has unique factorisation. An ordinary prime $p \equiv 2 \pmod 3$ remains prime here; in particular, **23 remains prime** [2].

Modulo 23, $x^2 + x + 1$ has no root: a root would be a nontrivial cube root of unity in the group $\mathbb{F}_{23}^\times$ of order 22, impossible since $3 \nmid 22$. Hence

$$\boxed{\mathbb{Z}[\omega]/(23) \cong \mathbb{F}_{23^2} = \mathbb{F}_{529}.}$$

Residues of $a + b\omega$ give $23 \times 23 = 529$ field elements.

Convention. The numerical values and magnitude comparison use principal logarithms. Other branches change the values; ω^η remains positive real on every branch.

Sources. [1] University of South Carolina, Math 785 notes, §8, Theorem 19. [2] Northeastern University, *Quadratic Integer Rings*, pp. 14, 31.